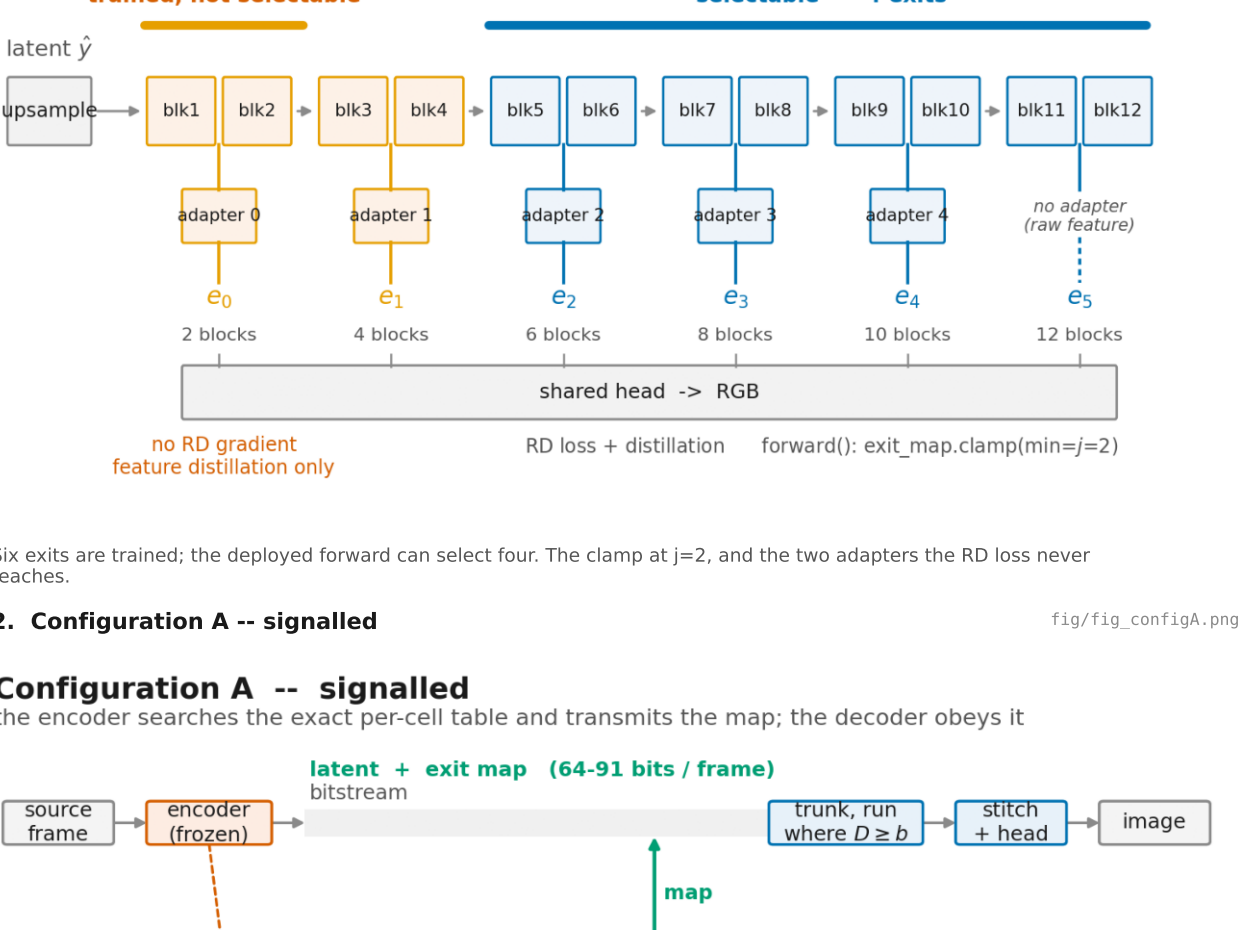


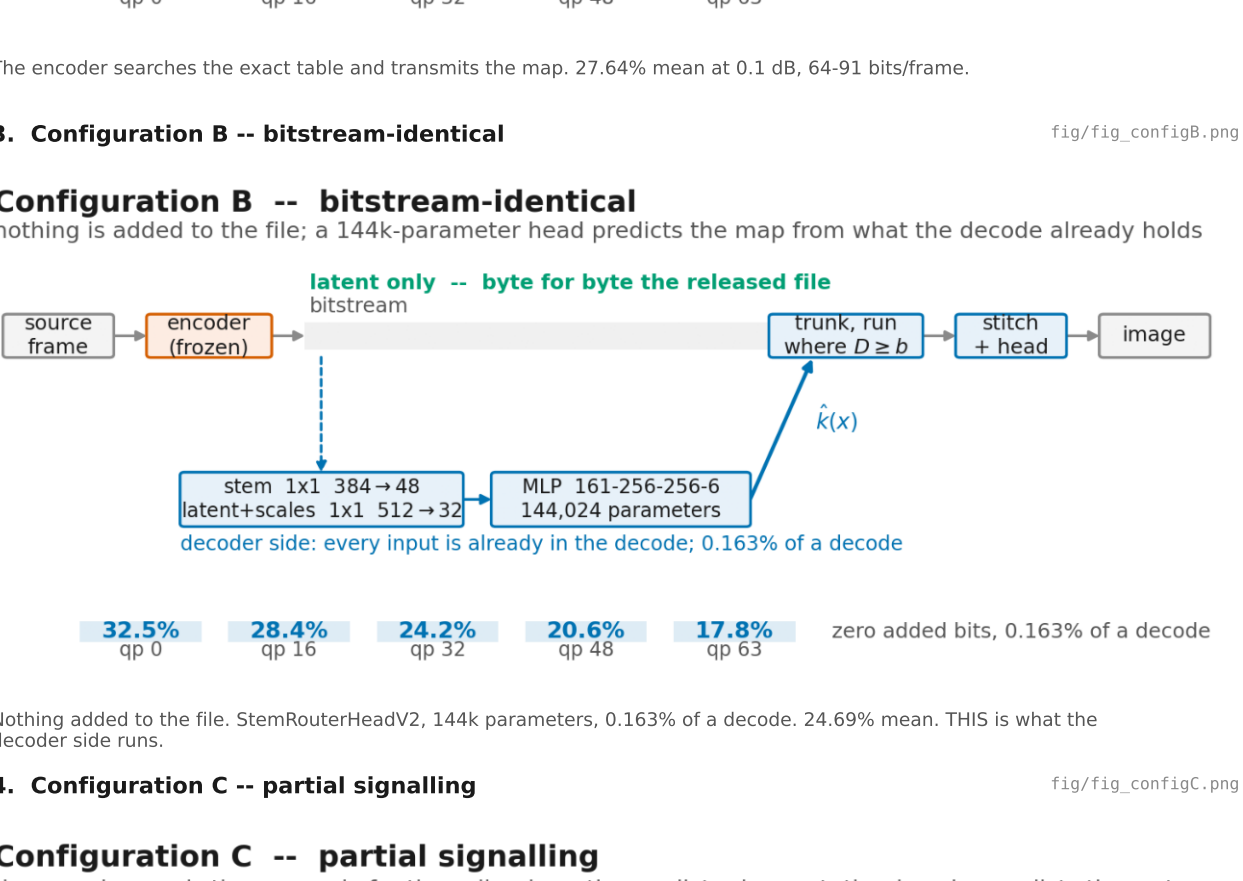
FLEX-UF -- every figure, in the order the argument runs

The ladder the main experiment trains fig/fig1_training.png



Six exits are trained; the deployed forward can select four. The clamp at $j=2$, and the two adapters the RD loss never reaches.

2. Configuration A -- signalled fig/fig_configA.png

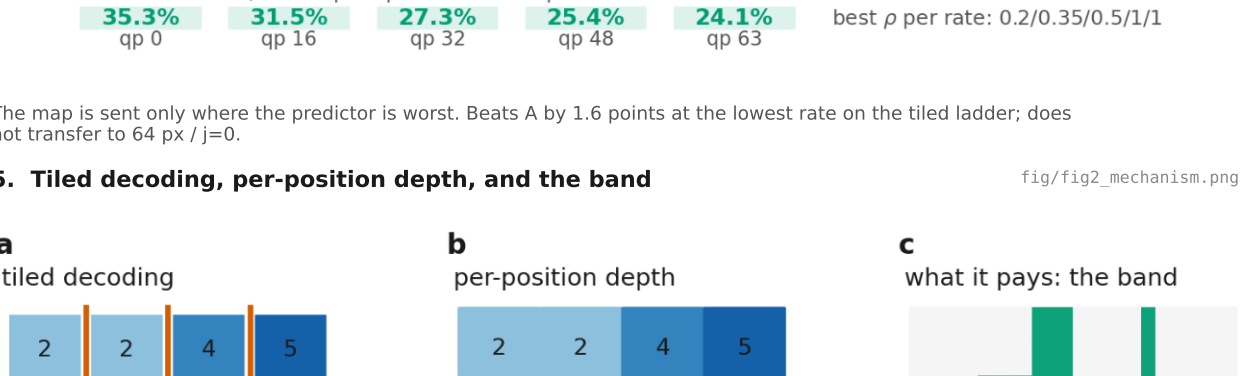


The encoder searches the exact table and transmits the map; the decoder obeys it

nothing is added to the file; a 144k-parameter head predicts the map from what the decoder already holds

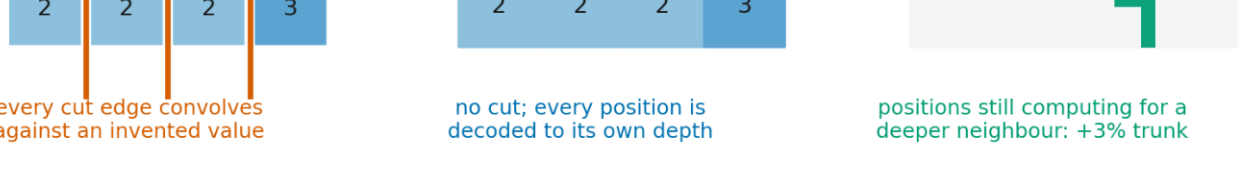
nothing is added to the file. StemRouterHeadV2, 144k parameters, 0.163% of a decode. 24.69% mean. THIS is what the decoder side runs.

3. Configuration B -- bitstream-identical fig/fig_configB.png



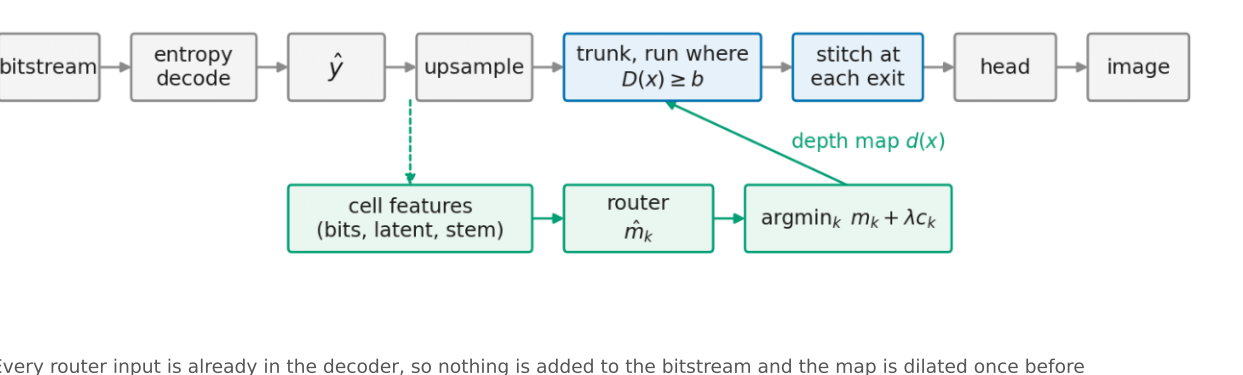
Nothing added to the file. StemRouterHeadV2, 144k parameters, 0.163% of a decode. 24.69% mean. THIS is what the decoder side runs.

4. Configuration C -- partial signalling fig/fig_configC.png



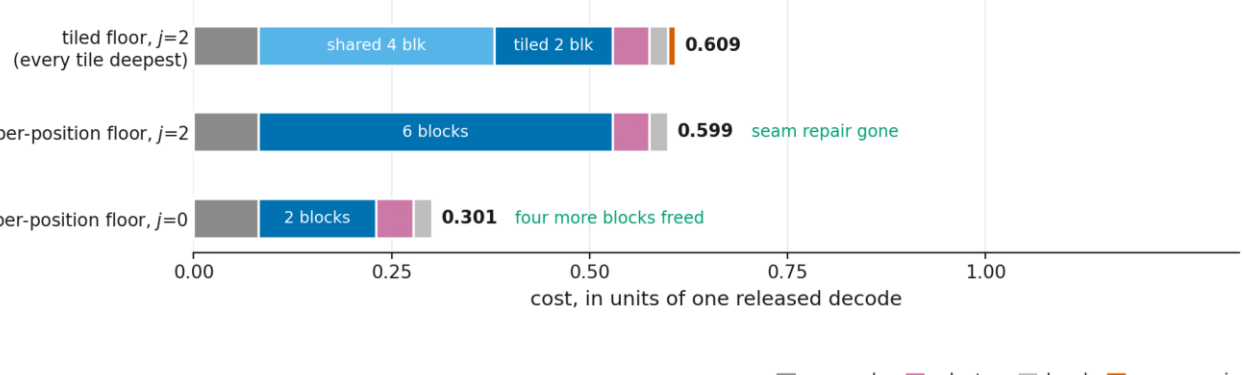
The map is sent only where the predictor is worst. Beats A by 1.6 points at the lowest rate on the tiled ladder; does not transfer to 64 px / $j=0$.

5. Tiled decoding, per-position depth, and the band fig/fig2_mechanism.png



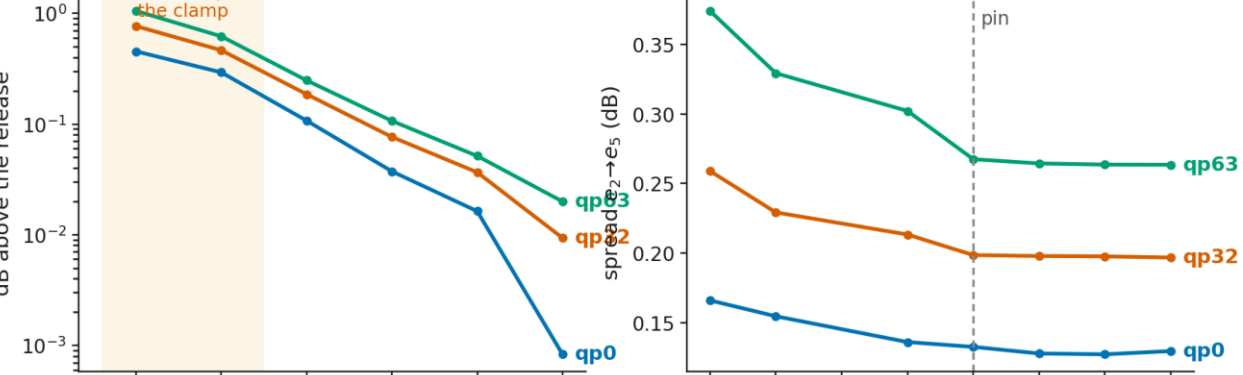
Every router input is already in the decoder, so nothing is added to the bitstream and the map is dilated once before the trunk runs.

6. The decoder with the allocation path drawn in fig/fig10_pipeline.png



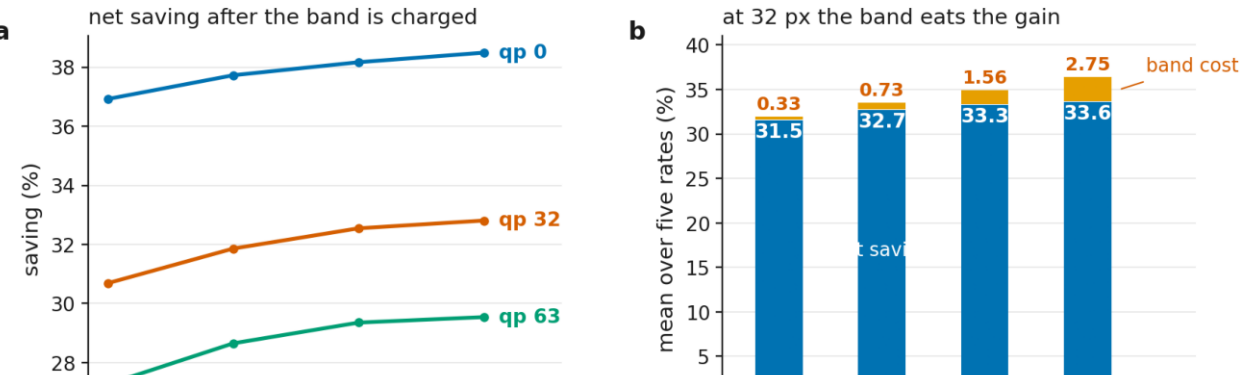
Every router input is already in the decoder, so nothing is added to the bitstream and the map is dilated once before the trunk runs.

7. Where one decode's compute goes fig/fig11_budget.png



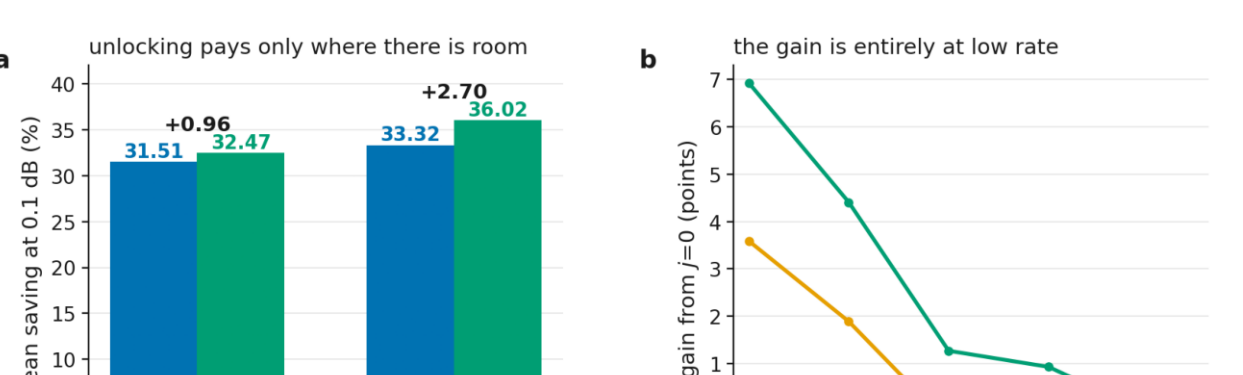
The floor caps the ceiling: 0.609 tiled, 0.599 per-position, 0.301 once the clamp stops forcing four shared blocks on every position.

8. The measured cost of each rung, and the spread by epoch fig/fig3_ladder.png



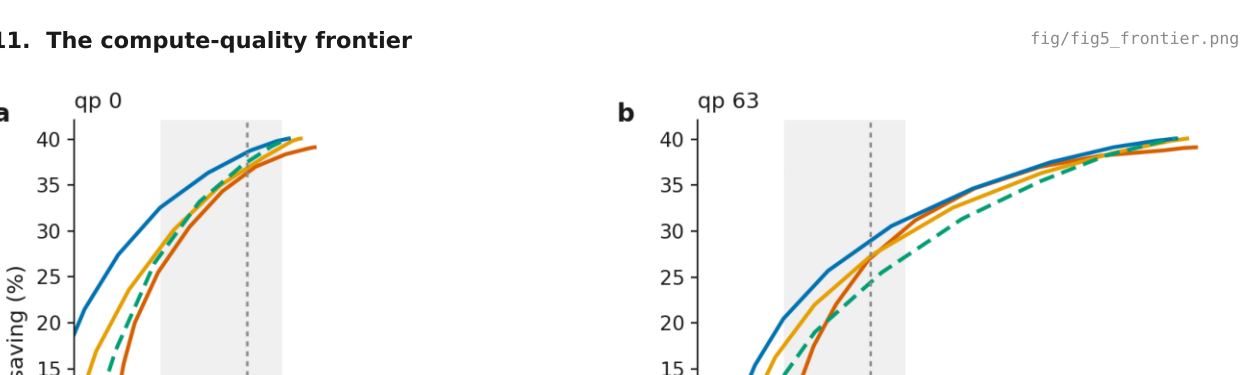
What the clamp closes off, and the fact that the ladder's spread stops improving at the pinned epoch.

9. Allocation cell against saving fig/fig4_granularity.png



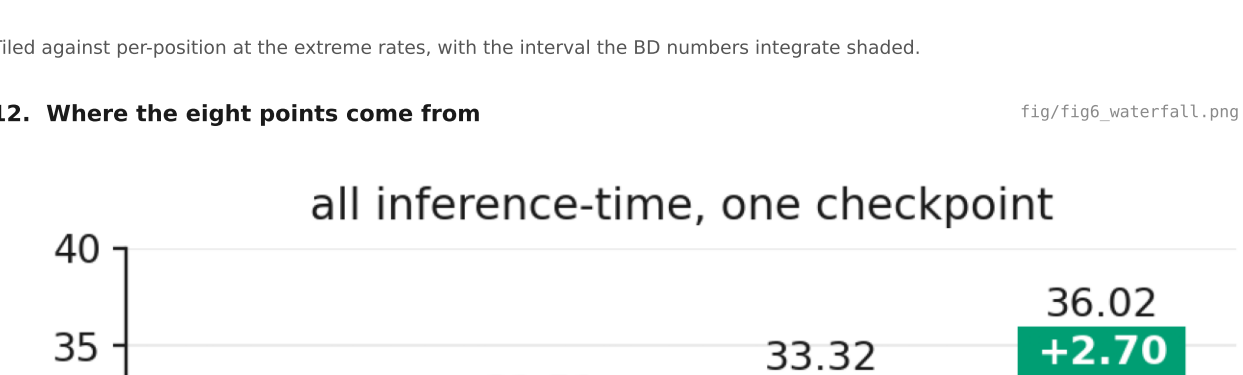
Finer cells buy saving and pay a dilation band; at 32 px the band eats the gain, which is why 64 px is the operating choice.

10. The two levers are not additive fig/fig12_interaction.png



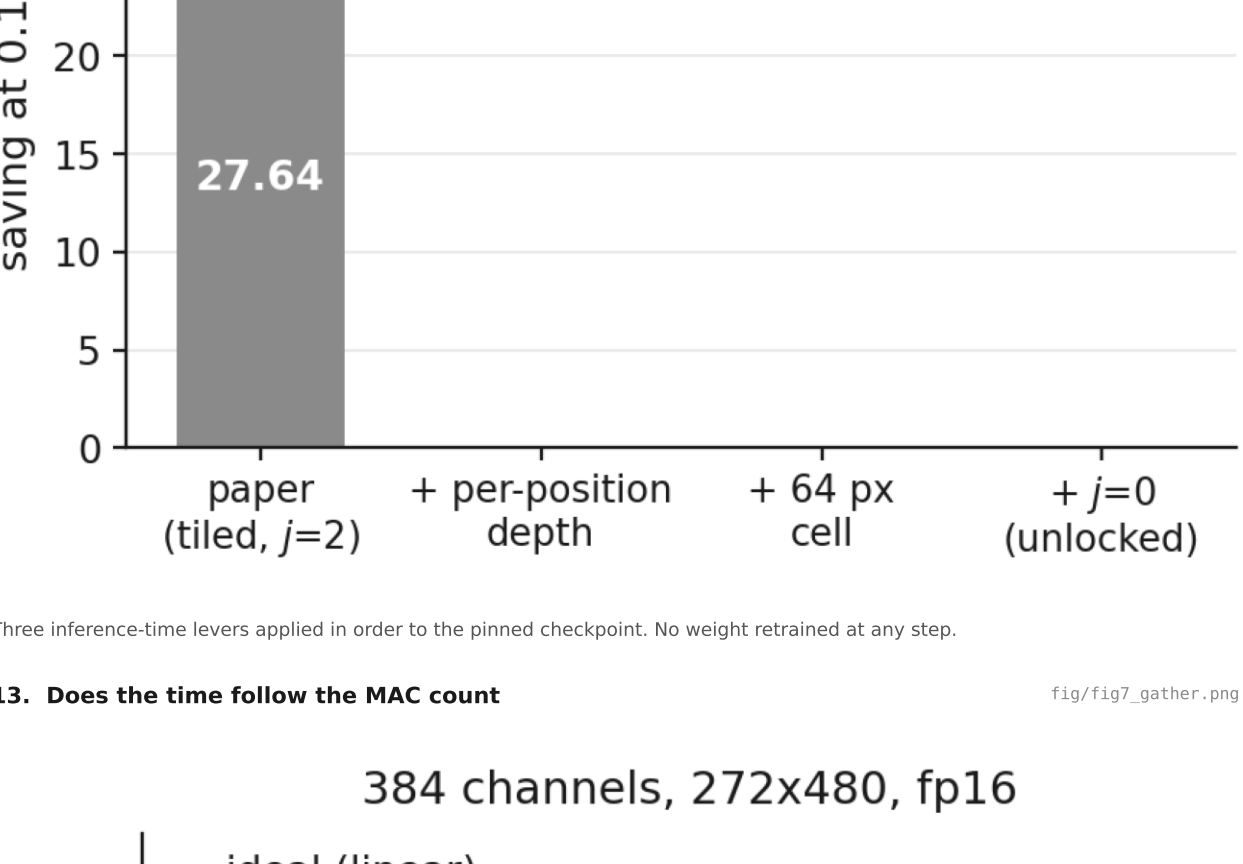
Unlocking the clamp is worth +0.96 points at 256 px and +2.70 at 64 px; the gain is entirely at low rate.

11. The compute-quality frontier fig/fig5_frontier.png



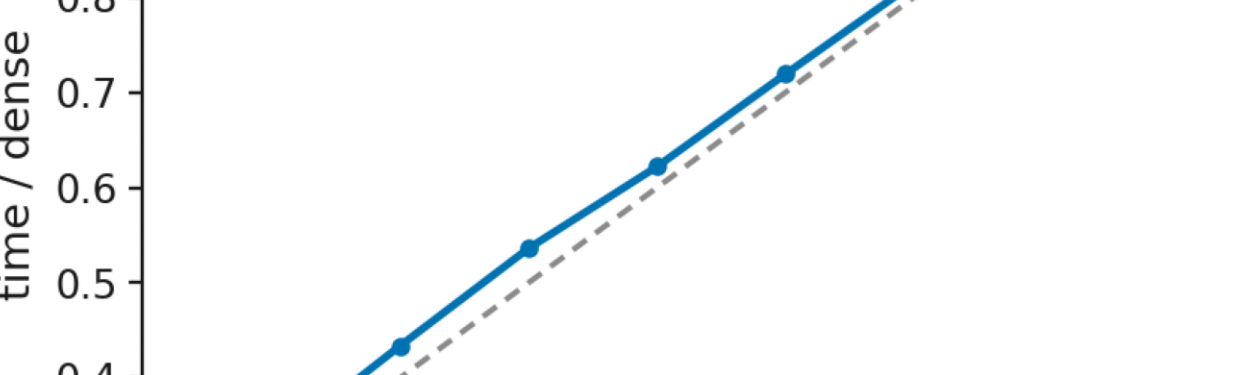
Tiled against per-position at the extreme rates, with the interval the BD numbers integrate shaded.

12. Where the eight points come from fig/fig6_waterfall.png



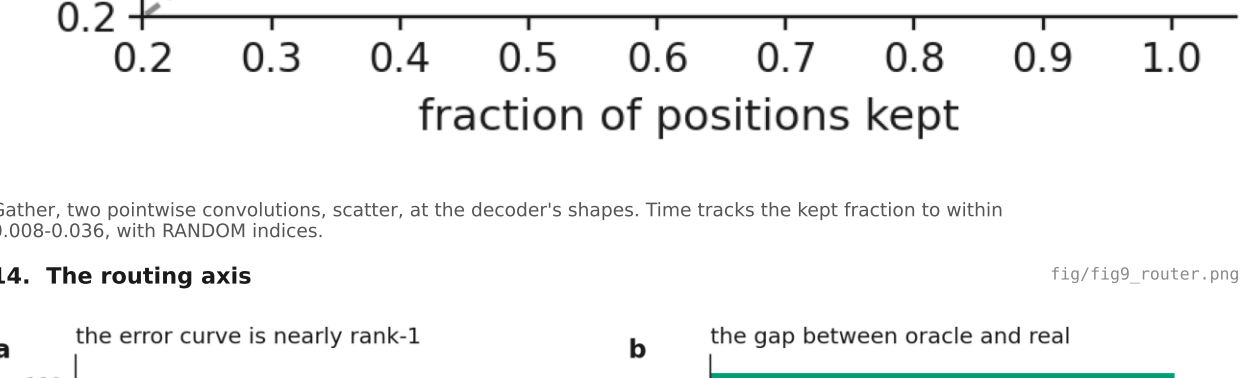
Three inference-time levers applied in order to the pinned checkpoint. No weight retrained at any step.

13. Does the time follow the MAC count fig/fig7_gather.png



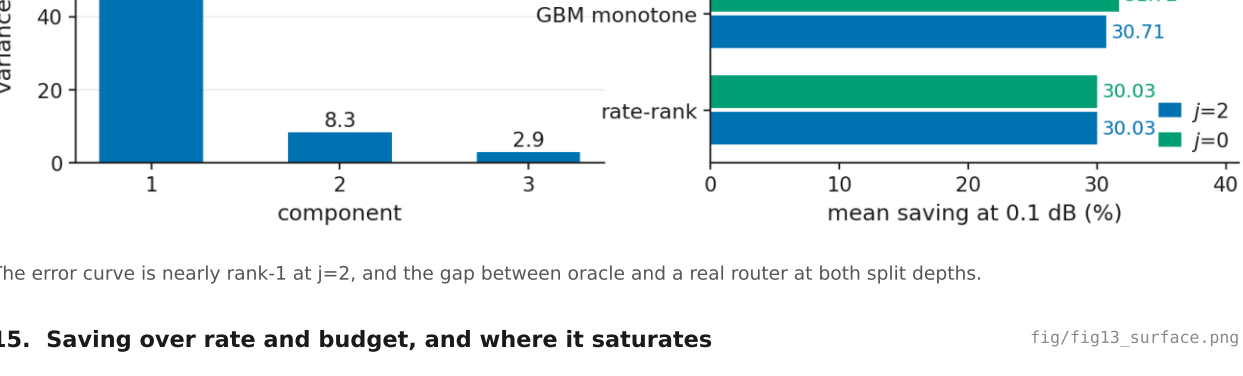
Gather, two pointwise convolutions, scatter, at the decoder's shapes. Time tracks the kept fraction to within 0.008-0.036, with RANDOM indices.

14. The routing axis fig/fig9_router.png



The error curve is nearly rank-1 at $j=2$, and the gap between oracle and a real router at both split depths.

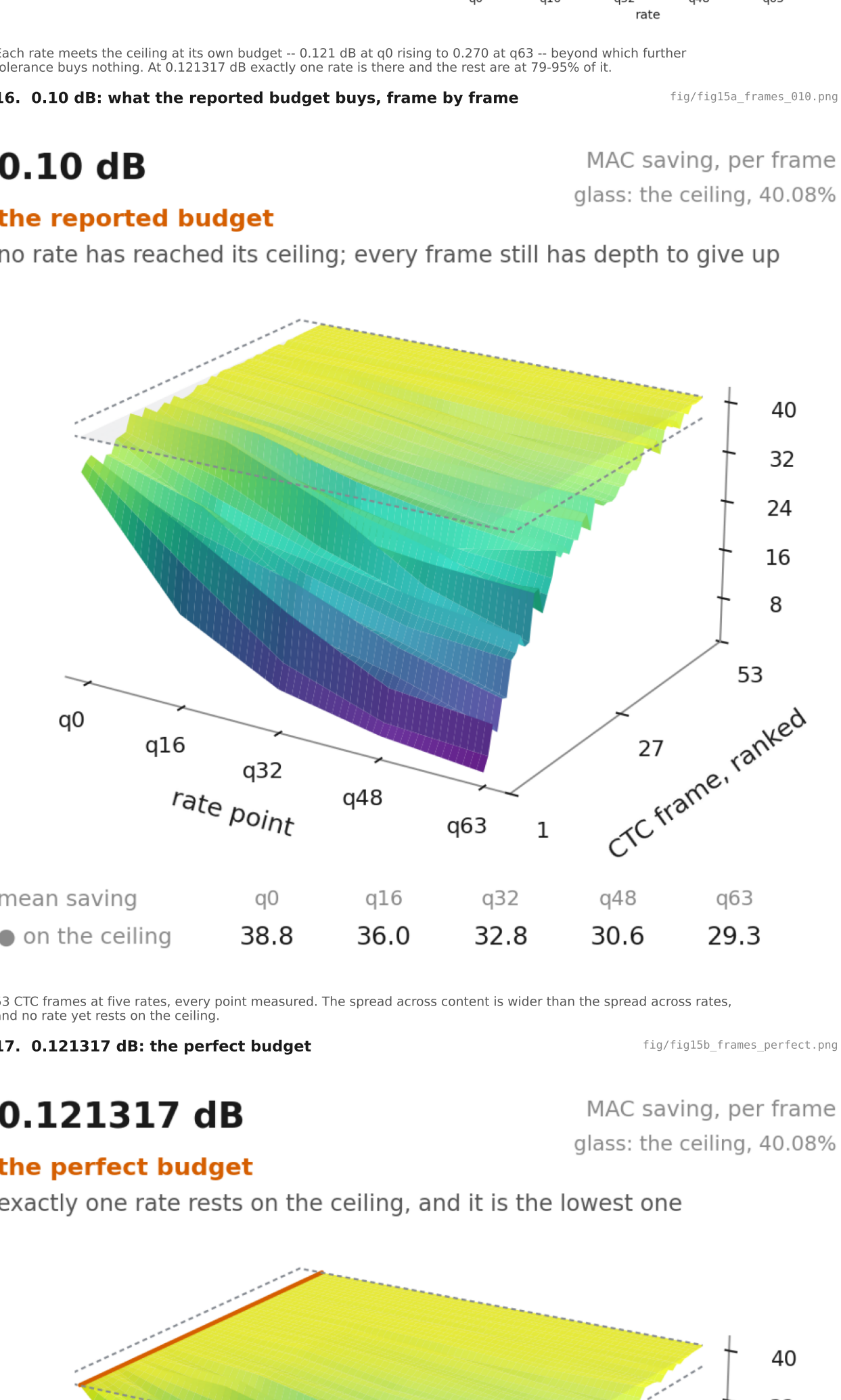
15. Saving over rate and budget, and where it saturates fig/fig13_surface.png



Each rate meets the ceiling at its own budget -- 0.121 dB at q0 rising to 0.270 at q63 -- beyond which further tolerance buys nothing. At 0.121317 dB exactly one rate is there and the rest are at 79-95% of it.

16. 0.10 dB: what the reported budget buys, frame by frame fig/fig15a_frames_010.png

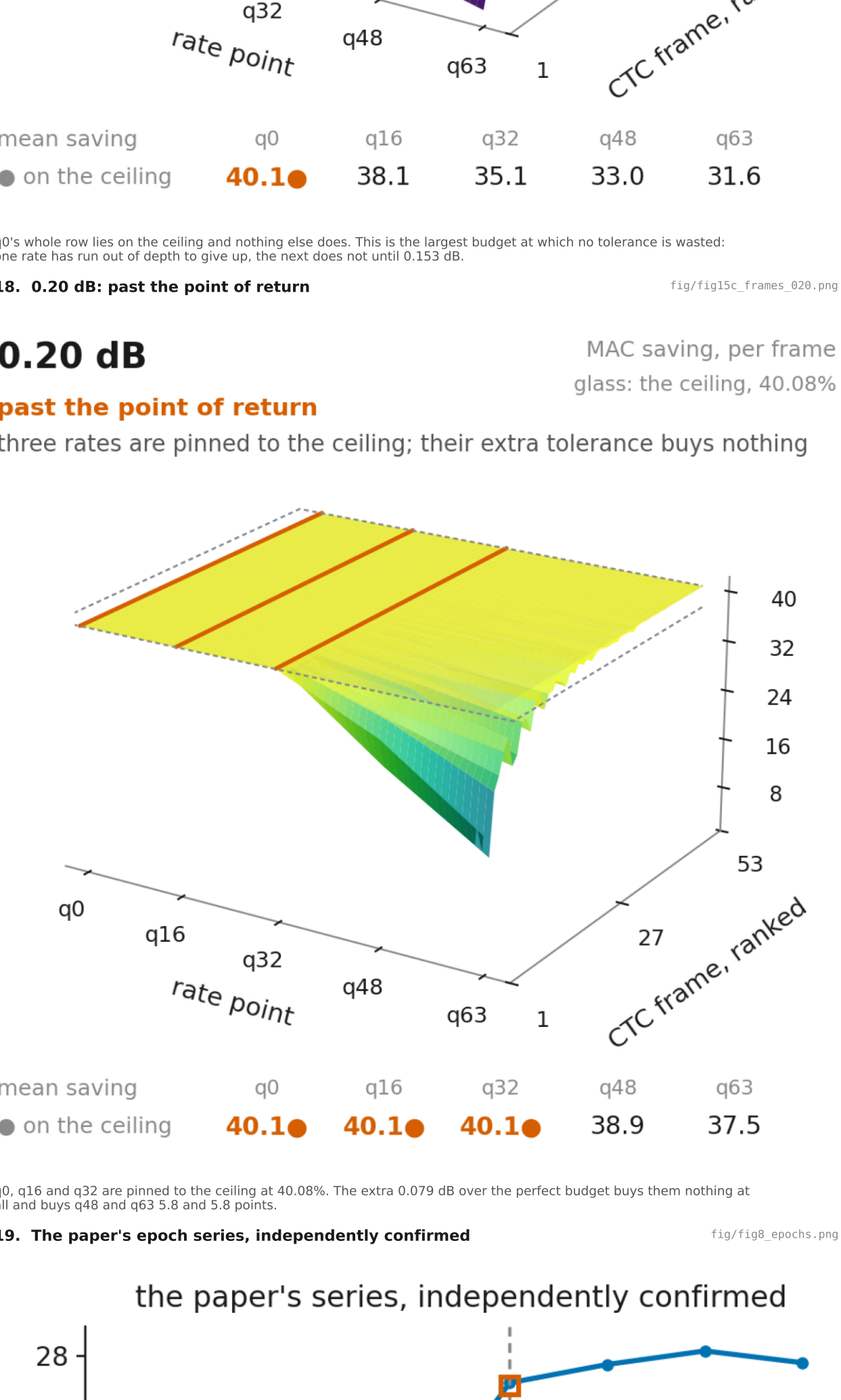
0.10 dB
the reported budget
no rate has reached its ceiling; every frame still has depth to give up



53 CTC frames at five rates, every point measured. The spread across content is wider than the spread across rates, and no rate yet rests on the ceiling.

17. 0.121317 dB: the perfect budget fig/fig15b_frames_perfect.png

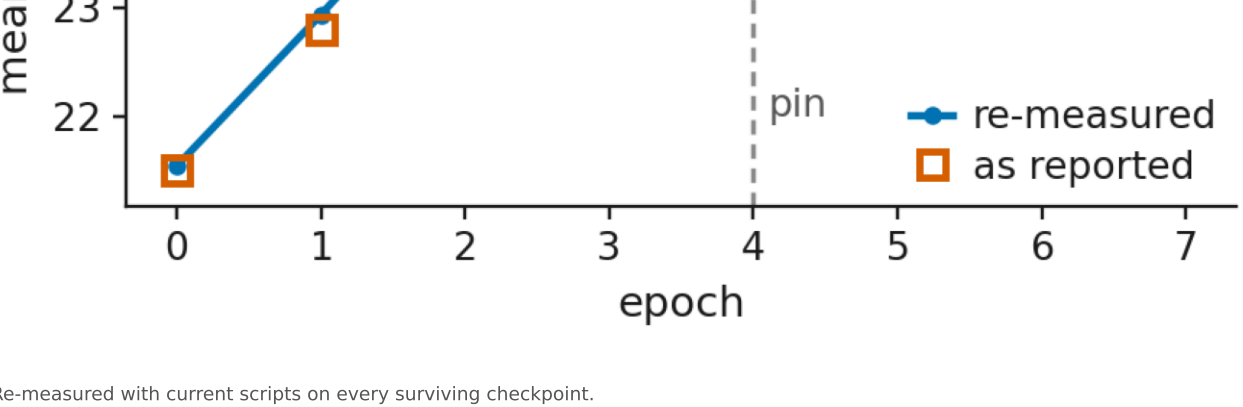
0.121317 dB
the perfect budget
exactly one rate rests on the ceiling, and it is the lowest one



q0's whole row lies on the ceiling and nothing else does. This is the largest budget at which no tolerance is wasted: one rate has run out of depth to give up, the next does not until 0.153 dB.

18. 0.20 dB: past the point of return fig/fig15c_frames_020.png

0.20 dB
past the point of return
three rates are pinned to the ceiling; their extra tolerance buys nothing



q0, q16 and q32 are pinned to the ceiling and nothing else does. This is the largest budget at which no tolerance is wasted: one rate has run out of depth to give up, the next does not until 0.153 dB.

19. The paper's epoch series, independently confirmed fig/fig8_epochs.png

the paper's series, independently confirmed

Re-measured with current scripts on every surviving checkpoint.